

PXRD-Diff: diagnosing the encoder bottleneck in diffusion-based powder-diffraction structure inversion

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Synopsis: A deliberately small conditional diffusion model for powder-diffraction structure inversion is used as a diagnostic instrument. The evidence points to global pooling in the pattern encoder as the stage that discards the absolute d-spacing information lattice recovery needs. A classical autoindexer, supplied with the true crystal system, serves as a control; three external generative models are re-scored on one shared evaluation harness; and a code audit prompted by review corrects three implementation errors.

Abstract

Inverting a one-dimensional powder X-ray diffraction (PXRD) pattern to a three-dimensional crystal structure remains an open problem. Recent generative approaches report results under mutually incompatible protocols, and their failure modes are seldom catalogued. This paper is a diagnostic study. PXRD-Diff, a 3.7 M-parameter conditional diffusion model trained on the CDVAE MP-20 split, is kept deliberately small so that its failure modes are visible; it is not proposed as a competitive inversion tool. DiffractGPT, PXRDnet and deCIFer are re-scored through the same pymatgen StructureMatcher harness and reach 18.9 %, 30.0 % and 73.8 % match respectively; PXRD-Diff reaches 1.6 %. The central finding is mechanistic. The learned PXRD encoder reaches a 0.007 normalised auxiliary lattice-regression loss in isolation, yet substituting that head’s own lattice into the sampler recovers only 1.2 % of structures, because its ~ 1.1 Å length error shifts every Bragg peak by ~ 15 %. Stronger, constrained supervision of the head does not reduce that error. The evidence therefore points to the global pooling that discards absolute peak positions, rather than to the denoiser alone [PENDING C2: one sentence with the controlled pooling-ablation outcome]. A perturbation study locates a sharp sensitivity knee at ~ 0.5 Å cell error. A true-lattice oracle lifts match to 4.5 % at $n = 1000$ (three seeds; 95 % Wilson confidence interval [3.8, 5.3]). As a classical control, a Q-space autoindexer that is supplied with the true crystal system replaces the learned lattice head and reaches 1.6 % [1.2, 2.1]. A three-seed paired McNemar test shows the gain over the learned head is significant ($p < 10^{-4}$; 27 fixed, none broken) and confined to the high-symmetry systems below the knee. A code audit prompted by review found and corrected three implementation errors, including an atomic form factor evaluated in the wrong convention.

The corrected simulator agrees with pymatgen at a median Pearson correlation of 0.999 over 1000 structures, and the affected runs were repeated [PENDING C1: one clause stating whether the retrained headline numbers changed]. The contributions are the encoder-bottleneck diagnosis, a four-item failure catalogue, the shared evaluation harness and the audited, released code.

Keywords: powder diffraction; autoindexing; crystal structure determination; conditional diffusion models; machine learning; differentiable simulation; reproducibility

1. Introduction

PXRD is the most common structural-characterization measurement in solid-state chemistry. The forward problem is solved by 1920s structure-factor physics and a one-line `pymatgen` call. The inverse is brutal: symmetry-equivalent structures produce identical Bragg reflections, intensities are orientation-averaged, and the 3D→1D reduction destroys atom labelling. The traditional indexing → Pawley/Le Bail → Rietveld pipeline needs a trained crystallographer and a starting model good enough to converge; for the long tail of new materials, structures stay unsolved.

Three recent generative approaches (DiffractGPT: Choudhary, 2025; Crystalyze: Riesel *et al.*, 2024; PXRDnet: Guo *et al.*, 2025) argue neural networks can shortcut this pipeline. They report match rates in the tens of percent, but under mutually incompatible evaluation protocols, and failure modes are seldom catalogued. A practitioner cannot easily form a calibrated expectation of *what is hard* about PXRD inversion.

We build a small reproducible conditional diffusion model (**PXRD-Diff**) and use it as a diagnostic instrument. Its size is a design choice: a 3.7 M-parameter model exposes an encoder bottleneck that capacity would hide. It is not proposed as an alternative to the larger systems it is compared with. Contributions:

1. **A small, fully released baseline.** 3.7 M parameters, trains on a single RTX 5090 in ~1.5 h on CDVAE MP-20. A PyTorch port of the `pymatgen` structure-factor calculation, differentiable with respect to coordinates and lattice, supplies an auxiliary pattern-matching loss. The port introduces no new physics (§3.3).
2. **Encoder-bottleneck diagnosis (the primary contribution).** A perturbation study (§5.5) locates a sharp ~0.5 Å sensitivity knee. A true-lattice oracle lifts match to **4.5 %** at $n = 1000$ (three seeds; Wilson 95 % CI [3.8, 5.3]). The encoder reaches a 0.007 *normalised* aux lattice loss in isolation, but substituting that head’s lattice into the sampler recovers only 1.2 % (§5.6). Its ~1.1 Å error sits well above the knee. The evidence points to the pooled encoding, not merely the denoiser, as the stage that fails to deliver absolute d-spacings; a controlled pooling ablation tests this directly (§5.6) [PENDING C2: one-line outcome of the pooling ablation]. The 4.5 % oracle ceiling shows coordinates remain a co-bottleneck.
3. **A classical control.** A Q-space autoindexer, supplied with the true crystal system, replaces the learned lattice head at sampling time and reaches 1.6 % [1.2, 2.1] where the learned head recovers nothing. A three-seed paired McNemar test confirms the difference ($p < 10^{-4}$; 27 fixed, 0 broken). The gain appears only where the indexer’s cell error falls below the 0.5 Å knee, i.e. in the high-symmetry systems (§5.2, §5.5). We treat this as a sanity check on the diagnosis, not as a method: a fifty-year-old algorithm given extra prior information is expected to beat the learned head. An indexer run without the crystal system is reported alongside (§3.5) [PENDING C3: one-line outcome of the unknown-system run].

4. **Four-item failure catalogue.** Top-K Debye rerank, Debye-gradient DDIM guidance, Wyckoff-letter embeddings and a distance-matrix aux loss all fail at $n \geq 200$; the latter two combined (v14) underperform the bare ε baseline. The gradient-guidance failure agrees with independent work showing the PXRD loss landscape is too rough for gradient descent (Segal *et al.*, 2025). The Wyckoff result is read narrowly: the embedding carried a label, not a constraint (§5.4).
5. **Reproduced baselines and a shared harness.** DiffractGPT (Mistral-7B + LoRA), PXRDnet (CDVAE + XRD encoder) and deCIFer are re-scored on our harness: **18.9 %** match (DGpt, $n = 1000$), **30.0 %** (PXRDnet, $n = 20$) and **73.8 %** (deCIFer, $n = 298$). These numbers set the scale of the diagnostic model; they are not a claim of competitiveness. A candidate divergence between match rate and space-group recovery across two baselines is raised in §5.3 and §6 as a hypothesis that the $n = 20$ PXRDnet sample cannot establish. Crystalyze’s checkpoint download link is inactive; we cite but cannot reproduce.
6. **A code audit.** Review of the released code identified three implementation errors: the atomic form factor was evaluated in the wrong convention, the Debye loss in x_0 -residual mode was evaluated on the wrong tensor, and patterns the indexer could not index were silently given the true lattice. §5.7 reports each error, its effect on every published number, and what was re-run.

2. Related work

Diffusion for crystals. CDVAE (Xie *et al.*, 2022) introduced the MP-20 protocol; DiffCSP (Jiao *et al.*, 2023) used joint coord+lattice diffusion; MatterGen (Zeni *et al.*, 2025) scaled to millions of structures. PXRD-Diff borrows DiffCSP’s joint diffusion, restricted to the conditional setting where a 1D pattern conditions generation.

Neural PXRD inversion. DiffractGPT (Choudhary, 2025) is a fine-tuned Mistral-7B that decodes peak list + formula $\rightarrow$ CIF tokens. Crystalyze (Riesel *et al.*, 2024) uses CDVAE conditioned on an XRD transformer encoder with post-hoc symmetry filtering. PXRDnet (Guo *et al.*, 2025) is a CDVAE variant with *iterative latent-space gradient guidance* (~ 3.5 M decoder ops/material), published in *Nature Materials* in 2025. deCIFer (Johansen *et al.*, 2025) is a concurrent autoregressive-language-model approach that, like DiffractGPT, emits CIF tokens. All four are larger than PXRD-Diff and report results under bespoke evaluation pipelines; §5.3 re-scores DiffractGPT, PXRDnet, and deCIFer on a single shared **StructureMatcher** harness. Crystalyze’s checkpoint download link is “not yet active” (verified 2026-06-01).

Where the unit cell comes from. Recent approaches differ in whether the unit cell is generated jointly with the coordinates or supplied as a separate input. PXRDGen (Li *et al.*, 2025) generates lattice and coordinates jointly by default, but can accept an independently determined cell, from a cell network or from conventional indexing, as an additional condition. XtalNet (Lai *et al.*, 2025) is an end-to-end equivariant approach, reported on metal–organic-framework benchmarks. Parackal *et al.* (2024) enumerate symmetry-confined Wyckoff arrangements and rank them with a learned energy model before fitting the pattern. XRDSol (Yu *et al.*, 2026) denoises coordinates with an equivariant diffusion model conditioned on the pattern, and takes the stoichiometry and the unit cell as inputs. That last design choice is the one this paper examines from the other side: XRDSol assumes an indexed cell, and the cell is exactly what our learned head fails to recover. Chitturi *et al.* (2021) regress lattice parameters from simulated patterns with a convolutional network and analyse

how impurities, noise and broadening degrade the prediction, a diagnostic aim close to ours. None of these is reproduced here. They are cited to place PXRDiff, a diagnostic instrument rather than a competing solver, within the range of current designs.

Classical structure solution from powder data. Recovering a unit cell from peak positions is a solved sub-problem with a fifty-year toolchain: the de-Wolff dichotomy method, ITO, TREOR and DICVOL (Boultif & Louër, 2004) index high-symmetry powder patterns in CPU milliseconds, with the difficulty concentrated in low-symmetry (monoclinic and triclinic) cells. Indexing is only the first stage of classical structure solution. Once the cell and space group are fixed, direct-space search, simulated annealing and charge flipping complete the structure, as implemented in programs such as FOX (Favre-Nicolin & Černý, 2002) and EXPO (Cuocci *et al.*, 2022). §3.5 reuses only the indexing stage, as a control for the learned lattice head. The indexer is given the true crystal system, which classical practice must itself infer. §5.2 and §5.5 show the resulting lift tracks the symmetry-dependent accuracy classical indexers have always had.

Differentiable physics. Scattering-pattern losses date to Rietveld. The module used here is a PyTorch port of the `pymatgen` structure-factor computation, differentiable with respect to coordinates and lattice so that it can serve as a diffusion-training loss term; it adds no physics beyond that computation (§3.3). Concurrent generative-inversion work conditions on differentiable spectral targets for amorphous and nanostructured systems (Guo & Schwalbe-Koda, 2026). Relevant to our negative results, Segal *et al.* (2026) show the powder-XRD similarity loss landscape is “too rough for gradient descent”, independent evidence for why our sampling-time Debye-gradient guidance fails (§5.4).

Equivariance. Not strictly $E(3)$ -equivariant: we use periodic-distance RBFs (SchNet-style; Schütt *et al.*, 2017). A frozen MACE (Batatia *et al.*, 2022) encoder showed no measurable benefit at MP-20 scale in pilots and was deferred.

3. Method

3.1 Problem formulation

Let $\mathcal{C} = (\mathbf{F}, \mathbf{Z}, \mathbf{L})$ denote a crystal structure with fractional coordinates $\mathbf{F} \in [0, 1)^{N \times 3}$, atomic numbers $\mathbf{Z} \in \{1, \dots, 100\}^N$, and a 3×3 lattice matrix $\mathbf{L}$ parametrised by $(a, b, c, \alpha, \beta, \gamma) \in \mathbb{R}^6$. Let $\mathbf{p}(\mathcal{C}) \in \mathbb{R}^{4251}$ be the simulated Cu $K\alpha$ PXRDiff pattern on a fixed 2θ grid from 5° to 90° at 0.02° resolution, normalised to maximum intensity 1.

We are given $\mathbf{p}$ and $\mathbf{Z}$ at test time, and we wish to sample $(\mathbf{F}, \mathbf{L}) \sim p(\cdot | \mathbf{p}, \mathbf{Z})$. Note that the composition $\mathbf{Z}$ being known is a meaningful simplification (in practice one usually knows the chemistry from synthesis), but it is consistent with prior work on the same task.

3.2 Architecture overview

PXRDiff has three trained components: a PXRDiff encoder, a denoiser, and a small auxiliary lattice head used only during training.

PXRDiff encoder. A 1D ResNet of four blocks (channels $64 \rightarrow 128 \rightarrow 256 \rightarrow 256$, stride-2 downsampling, GroupNorm, SiLU) ingesting the standardised pattern. We expose two outputs: a global pooled vector $\mathbf{g} \in \mathbb{R}^{256}$ and a multi-resolution feature map $\mathbf{F}_{\text{pxrd}} \in \mathbb{R}^{L \times 256}$ obtained by

1×1-projecting each block output to d_{model} and concatenating along the spatial axis. *The multi-resolution map is critical.* In an early run (referred to as `gpu_v4` in our logs) we conditioned the denoiser on $\mathbf{g}$ only via additive broadcast, and the coordinate loss never moved off the random baseline of 3.0; the auxiliary head told us the encoder was learning useful features (its loss dropped from 0.99 to 0.007), but the `AdaptiveAvgPool1d(1)` collapse destroyed all spectral structure before it could reach the denoiser.

Denoiser. A periodic-distance message-passing network with $L = 3$ layers and $d_{\text{model}} = 256$. Atom embeddings $\mathbf{h}_i^{(0)} = \text{Emb}(\mathbf{Z}_i) + W_{\text{coord}}\mathbf{F}_i^{(t)}$ are updated by alternating (i) message passing over RBF-encoded periodic Cartesian distances under minimum-image convention, with timestep FiLM conditioning, and (ii) cross-attention from atoms (queries) to multi-resolution PXRD features (keys/values). Two output heads predict per-atom coordinate noise $\epsilon_F \in \mathbb{R}^{N \times 3}$ and lattice noise $\epsilon_L \in \mathbb{R}^6$. The lattice head is a small MLP applied to $[\bar{\mathbf{h}}; \mathbf{g}; W_{\text{lat}}\ell^{(t)}; \mathbf{t}_{\text{cond}}]$: that is, the pooled atom features, the global PXRD embedding, **a projection of the noisy lattice itself**, and the timestep encoding (see §3.4).

Auxiliary head. A single-hidden-layer MLP that predicts $(a, b, c, \alpha, \beta, \gamma)$ from $\mathbf{g}$ alone, trained with an MSE loss against the (normalised) ground-truth lattice. Its purpose is purely diagnostic, to expose whether the encoder is learning any lattice-relevant representation, and its loss does not flow back into the denoiser.

3.3 Differentiable Bragg structure-factor loss

This module is an engineering component, not a physical contribution. It is a PyTorch port of the structure-factor calculation in `pymatgen.analysis.diffraction.xrd.XRDCalculator`, written so that gradients flow to fractional coordinates and lattice. We implement

$$F(hkl) = \sum_j f_j(s) \exp(-B_{\text{iso}}s^2) \exp[2\pi i(hx_j + ky_j + lz_j)],$$

with the atomic form factor in the convention `pymatgen` uses for its tabulated coefficients,

$$f_j(s) = Z_j - 41.78214s^2 \sum_{k=1}^4 a_k \exp(-b_k s^2),$$

where (a_k, b_k) are read from `ATOMIC_SCATTERING_PARAMS`, Z_j is the atomic number, $s = \sin \theta / \lambda$, and $B_{\text{iso}} = 0.5 \text{ \AA}^2$ is a uniform isotropic temperature factor. The intensity at each reflection is $|F(hkl)|^2$ multiplied by the standard Lorentz-polarisation correction. We enumerate all (h, k, l) with $|h|, |k|, |l| \leq 5$ (1330 reflections after excluding the origin), place each on the 2θ grid via Bragg’s law, and broadcast each reflection as a Gaussian peak with FWHM 0.1° to obtain a continuous, differentiable pattern on a 256-bin coarse grid. The module treats ideal powders only: no preferred orientation, peak asymmetry, instrument profile or background is modelled. It is a single `nn.Module`, vectorised over the batch dimension.

Correction made during review. The submitted version of this paper evaluated the form factor as the bare sum $\sum_k a_k \exp(-b_k s^2)$ with the same coefficients. That expression is not the one the coefficients were fitted for. It gives the wrong element weighting: for silicon it returns $f(0) = 5.8$ where the correct value is $Z = 14$, and it decays too quickly with s . Both referees who inspected the code identified the discrepancy. The module was corrected to the expression above and re-validated (below). The training runs that used the Debye loss were repeated with the corrected

module [PENDING C1: state which runs were retrained and whether the numbers in Tables 1 and 2 changed, and by how much].

Validation. On the first 1000 MP-20 test structures we compute the Pearson correlation between the differentiable pattern and the `pymatgen.XRDCalculator` reference, both on the 256-bin coarse grid. With the corrected form factor the mean is 0.988 and the median 0.999; 98.0 % of structures exceed 0.9. With the submitted version’s form factor the same structures give a mean of 0.952 and a median of 0.969; 90.7 % exceed 0.9. The correction improves the agreement for every one of the 1000 structures. On the first 50 structures the means are 0.939 (submitted form factor) and 0.974 (corrected). The submitted version reported 0.962 on 50 structures with the validation script’s default reflection range ($|h|, |k|, |l| \leq 10$), which is wider than the training configuration (≤ 5) used for the numbers above. Re-running that wider range reproduces the submitted value for the submitted expression (0.963 on the first 50 structures, 0.960 on all 1000) and gives 0.999 for the corrected expression at both sizes (Appendix C). The wider range also removes most of the lower tail, which is therefore a truncation effect of the ≤ 5 training grid on small cells rather than a property of the form factor. Gradients through the structure factor with respect to fractional coordinates were verified non-zero.

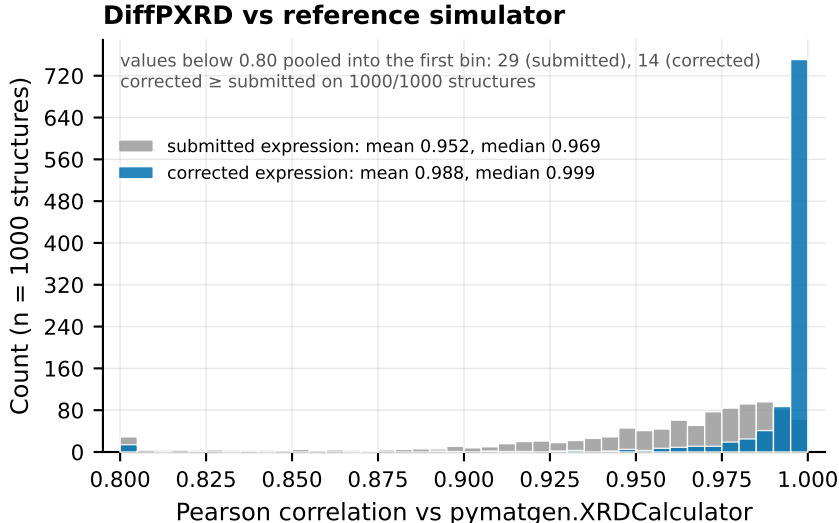


Figure 1: **Figure 1. Differentiable-simulator validation.** Distribution of Pearson correlation between the PyTorch DiffPXR module and `pymatgen.XRDCalculator` over the first 1000 MP-20 test structures, with the corrected form factor (mean 0.988, median 0.999) and with the form factor of the submitted version (mean 0.952, median 0.969). The correction improves every structure. The residual lower tail is dominated by layered structures with strong texture in the reference.

As a loss. During training we recover an estimate $\hat{\mathbf{F}}$ of the clean coordinates from the model output (see §3.4), simulate $\hat{\mathbf{p}} = \text{DiffPXR}(\hat{\mathbf{F}}, \mathbf{Z}, \mathbf{L}_{\text{true}})$, and compare with the input pattern via a Pearson-correlation loss $\mathcal{L}_{\text{Debye}} = 1 - \rho(\hat{\mathbf{p}}, \mathbf{p})$. We use the true lattice for this auxiliary loss because the differentiable simulator is more sensitive to lattice errors than to coordinate errors at the early stages of training, and disentangling the two signals proved more stable.

3.4 Training objective

We use a VP-SDE with cosine schedule (Nichol & Dhariwal, 2021) for both channels:

$$\bar{\alpha}(t) = \cos^2\left(\frac{\pi}{2} \cdot \frac{t+s}{1+s}\right), \quad s = 0.008, \quad t \in [0, 1].$$

Coordinates are diffused on the flat torus $\mathbb{T}^3$ by wrapping the noisy sample to $[0, 1)^3$; the loss is computed on the periodic difference $(\hat{\mathbf{F}} - \mathbf{F} + 0.5) \bmod 1 - 0.5$. Lattice parameters are diffused in $\mathbb{R}^6$ after standardisation by the train-set mean and standard deviation.

The full training objective (best configuration, run `gpu_v13`) is

$$\mathcal{L} = \mathcal{L}_{\text{coord}} + \lambda_{\text{lat}} \mathcal{L}_{\text{lat}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{Debye}} \mathcal{L}_{\text{Debye}},$$

with $\lambda_{\text{lat}} = 0.1$, $\lambda_{\text{aux}} = 0.5$, $\lambda_{\text{Debye}} = 1.0$.

Two non-trivial parameterisation choices. Both are essential and were arrived at by ablation, not foresight.

x_0 -residual prediction. Rather than asking the model to predict noise ϵ , the heads predict a **residual** that is added to the noisy input to obtain the clean estimate:

$$\hat{\mathbf{F}}^{(0)} = \mathbf{F}^{(t)} + \text{Coord-Head}(\mathbf{h}), \quad \hat{\ell}^{(0)} = \ell^{(t)} + \text{Lat-Head}(\cdot).$$

Because the heads are MLPs initialised to output near-zero, this means the model starts as the identity transform: at $t \approx 1$ (near pure noise) the model outputs the noisy input as its $\hat{\mathbf{F}}^{(0)}$ estimate, which is wrong but at least bounded; ϵ -prediction with the same architecture had to learn the entire transform from random initialisation and never recovered. At single seed this change appeared to lift the headline match metric 3.5 $\times$; a multi-seed checkpoint sweep (§7) shows that match-rate lift does not replicate, but the parameterisation does reproducibly improve the pattern-Pearson ($\approx 0.36 \rightarrow 0.40$, see §5.1).

Correction made during review. In x_0 -residual mode the Debye loss must be evaluated on the clean-coordinate estimate $\hat{\mathbf{F}}^{(0)} = \mathbf{F}^{(t)} + \text{Coord-Head}(\mathbf{h})$. The submitted code evaluated it on the residual $\text{Coord-Head}(\mathbf{h})$ alone. For every x_0 -residual run (v13, v14, v15, v16 and the Phase 9 checkpoint v21) the Debye term therefore compared the pattern of the wrong tensor with the target. The ϵ -mode runs v10 and v11 were unaffected, as were the coordinate and lattice losses and every evaluation-time number. The code is fixed. The affected runs were retrained with the corrected loss and the corrected simulator [PENDING C1: retrained numbers for the x_0 -residual runs].

Lattice-head input. The lattice head reads the noisy lattice $\ell^{(t)}$ as an explicit input feature. Without this, i.e. if the head only sees the (clean) lattice that is also passed to the denoiser as the geometry context, the head can only ever predict the unconditional mean noise, $\mathbb{E}[\epsilon] = 0$, and its loss stays pinned at the random baseline of 1.0 forever. With this fix, the lattice loss drops to 0.02–0.06 on stable runs.

We also experimented with two extensions that did not pan out (§5.4): a **Wyckoff-site embedding** added to atom features, and an auxiliary **distance-matrix loss** in which a pairwise MLP predicts ground-truth periodic distances between atoms. Both are documented in the released code behind the `--use-wyckoff` and `--dist-weight` flags.

3.5 Classical indexing as a control for the learned lattice head

The lattice head described in §3.2 is the *only* path by which the model encodes absolute d-spacings: every Bragg reflection 2θ position is set by the lattice through Bragg’s law, so the lattice parameters are the absolute reference frame against which all peak positions in the input pattern must be interpreted. As §5.5 shows, the encoder + denoiser learns *relative* d-spacing structure (Pearson 0.43 between predicted and target patterns at evaluation time, vs 0.97 between target and ground truth) but not the absolute scale. Classical autoindexing, which extracts peak positions, maps them to Q-space ($Q = 4\pi \sin\theta/\lambda$), and fits a unit cell by enumeration of $(h,k,l) \rightarrow Q^2(h,k,l)$, has solved this sub-problem since the 1970s and runs on a CPU in milliseconds per pattern.

We implement a from-scratch Q-space autoindexer that takes (i) the same simulated PXRD pattern fed to the encoder and (ii) the true crystal system of the target structure, supplied as a Bravais code, and returns a candidate $(a, b, c, \alpha, \beta, \gamma)$. Item (ii) is prior information beyond the composition that the learned head receives. It makes the indexer an optimistic control rather than a fair competitor, and every indexer number in this paper should be read with that in mind (see also §7 and the unknown-system run below). Peak picking uses a 1D local-maximum filter with a relative-intensity floor; the candidate Q-vector is fit by least squares to the de-Wolff dichotomy parameter form for each Bravais lattice. At sampling time we run the indexer first, substitute its output for the lattice channel of the diffusion sampler, and run DDIM only on the coordinate channel.

Per-system accuracy ($n = 1000$ MP-20 test). Native indexer overall: 48.8 % strict cell match, 1.45 Å mean cell-length MAE. Per crystal system: hexagonal 77.9 % / 0.53 Å, orthorhombic 59.8 % / 0.96 Å, trigonal 43.3 % / 4.62 Å, monoclinic 18.9 % / 1.76 Å, triclinic 0 %. The low-symmetry blowup (monoclinic and triclinic each have ≥ 4 free cell parameters and the de-Wolff search becomes hypothesis-capped) is the dominant remaining error. We ship a GSAS-II (Toby & Von Dreele, 2013) adapter (`src/pxrd_diff/indexer_gsas.py`) for the low-symmetry path behind a `--use-gsas` opt-in flag; in practice `DoIndexPeaks` hangs on real MP-20 monoclinic/triclinic patterns inside `findBestCell` and is documented experimental rather than headline.

Without the crystal system. At a referee’s request the indexer was also run with no crystal system supplied. Cubic, tetragonal, hexagonal/trigonal, orthorhombic and monoclinic cells are all searched, and the candidate with the highest de-Wolff M20 figure of merit is kept. [PENDING B2: per-system strict % and length MAE, and the fraction of patterns whose lattice type is recovered, on the same $n = 1000$.] [PENDING C3: downstream match rate at three seeds when these cells replace the learned head.]

The indexer is a drop-in in the mechanical sense: training is unchanged, and the diffusion sampler is unchanged except for the lattice-channel substitution at $t = T$. Patterns for which the indexer returns no cell (39 of the 1000 test patterns) are scored as misses. The submitted version had given them the true lattice; §5.7 reports the correction and its effect. The headline metric (§5.2) reports both with and without the indexer for direct comparison.

3.6 Sampling

DDIM (Song *et al.*, 2021) with 50 steps and $\eta = 0$. We start from independent standard Gaussian noise on the lattice and on the (un-wrapped) coordinates, then alternate the standard DDIM update on each channel. Coordinates are wrapped to $[0, 1)$ after every step. Predicted lattice parameters are de-standardised at the end and clipped to physically valid ranges $(a, b, c) \in [0.5, 100]$ Å, $(\alpha, \beta, \gamma) \in [10^\circ, 170^\circ]$ before being passed to `pymatgen.Lattice.from_parameters`.

A subtle bug fix is worth noting. In the x_0 -residual variant, recovering the implicit ε for the DDIM update requires dividing by $\sqrt{1 - \bar{\alpha}(t)}$; near $t = 0$ this denominator vanishes. We clamp it to 0.05 and skip the very last DDIM step when sampling, which removed a class of structures with NaN coordinates that we initially saw.

4. Experiments

Dataset. Canonical CDVAE (Xie *et al.*, 2022) MP-20 split: 27 136 / 9 047 / 9 046 train/val/test, ≤ 20 atoms/conventional cell. Patterns simulated with `pymatgen.XRD Calculator` (Cu $K\alpha_1$, $\lambda = 1.54184$ Å), max-normalised, 4 251 bins 5–90° / 0.02°; cached as `.npz`, zero failures over 45 196 structures.

Evaluation. Three views: (1) **composition match** (trivial, $\mathbf{Z}$ given); (2) **structure match** via `pymatgen.StructureMatcher` at $(\ell_{\text{tol}}, s_{\text{tol}}, \alpha_{\text{tol}}) = (0.2, 0.3, 5^\circ)$, the headline “match rate”; (3) **coordinate RMSD** on aligned, permuted atoms when matched. We also report space-group match at `symprec` $\in \{0.01, 0.05, 0.1, 0.2\}$, pattern Pearson, and R_{wp} .

The stricter “**all-correct**” rule combines `composition` $\wedge$ `sg-match@symprec=0.1` $\wedge$ `rmsd` ≤ 0.1 Å, the experimentally relevant case for downstream crystallographic use.

Two evaluation modes: *full pipeline* (predicted lattice + coords) and *true-lattice/coord-only* (ground-truth lattice substituted). The latter isolates coordinate quality from lattice quality and is the stricter ablation setting. In the indexer mode (§3.5) a pattern the indexer cannot index counts as a miss; it is never scored with the true lattice.

Implementation. AdamW, learning rate 10^{-3} (the value stored in the released v21 checkpoint; the submitted text gave 5×10^{-4} , see §5.7(d)), cosine decay to zero over 100 k steps, WD 10^{-4} , grad-clip 1.0. Batch 64, ~24 epochs, single RTX 5090 (~1.5–1.7 h/run). 3.7 M parameters; $L = 5, d = 384$ (~10 M) gave no benefit (§5.2). VP-SDE with $t \sim U(0, 1)$. Total compute: ~35 GPU-hours, ~USD 25 on Vast.ai (Phase 4 ablation + Phase 9 v21 retrain + indexer/perturbation sweeps + DGpt n=1000 + PXRdnet n=20 + evaluation).

5. Results

Order: Phase 4 architectural ablation (§5.1) $\rightarrow$ Phase 9 indexer drop-in (§5.2) $\rightarrow$ reproduced baselines (§5.3) $\rightarrow$ failure catalogue (§5.4) $\rightarrow$ encoder-bottleneck perturbation study (§5.5).

5.1 Phase 4: architectural ablation (true-lattice setting)

Table 1: seven runs sweeping the Debye loss, the lattice-input fix, x_0 -residual, and two extensions (Wyckoff, distance loss), all on $n = 1\,000$ MP-20 test with true lattice substituted. Every row is a single training seed. The table is exploratory: it guided the design, and only the ε -vs- x_0 -residual contrast was later replicated across seeds (§7). Runs v11 and v13–v16 were trained with the submitted version’s simulator, and v13–v16 with the x_0 -mode Debye defect (§3.3, §3.4) [PENDING C1: retrained rows, or a statement of which rows were retrained].

Table 1. Phase 4 coordinate-only ablation, MP-20 test ($n = 1\,000$, true lattice substituted). Single seed per row; exploratory.

Run	Parameterisation	λ _Debye	Wyckoff	λ _dist	Match %	Pearson	RMSD (Å)
v10	ε	0	—	0	1.40	0.359	0.17
v11	ε	1	—	0	0.90	0.365	0.15
v13	x_0-residual	1	—	0	2.51	0.434	0.22
v14	x_0 -residual	1	yes	0.01	0.80	0.367	0.14
v15	x_0 -residual	1	yes	0	2.10	0.392	0.21
v16	x_0 -residual	1	—	0.01	1.80	0.368	0.22

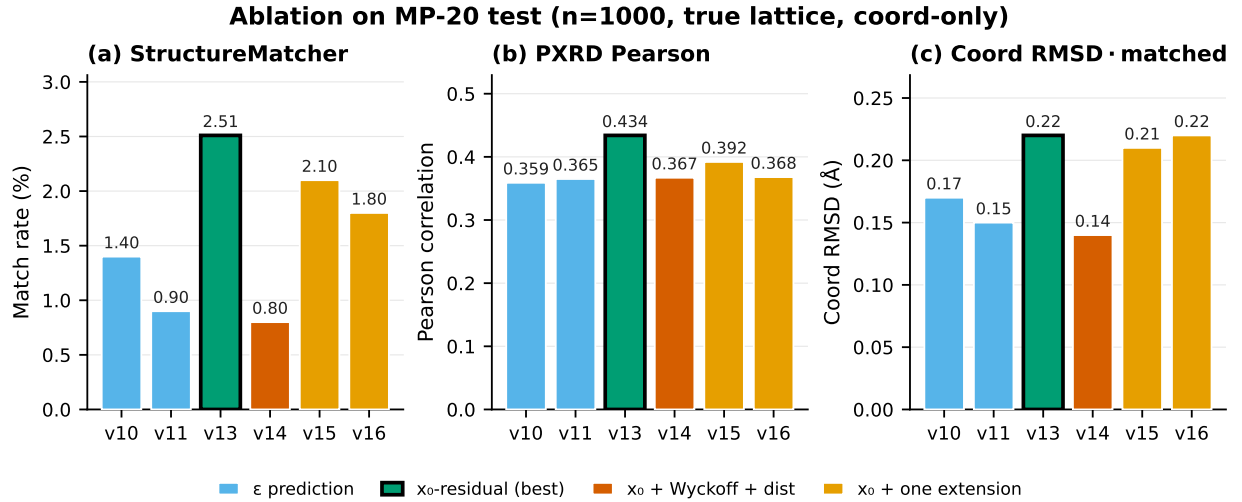


Figure 2. Phase 4 architectural ablation (single seed per run; exploratory). Match rate (a) and pattern Pearson (b) for the seven coordinate-only runs of Table 1 on MP-20 test ($n = 1000$, true lattice substituted). The $\varepsilon \rightarrow x_0$ -residual switch (v11 $\rightarrow$ v13) shows the largest single-seed match lift (panel a), but that match lift does not replicate across seeds; only the pattern-Pearson gain (panel b) does (§7); adding Wyckoff embeddings or a distance-matrix loss regresses below it.

The **lattice-input fix** (§3.4) is load-bearing: it drops lattice loss from 1.0 (pinned at the prior variance for 100 k steps) to ~ 0.02 once $W_{\text{lat}}\ell^{(t)}$ enters the head input. A second, apparent lift did **not** survive replication: switching from ε to x_0 -residual at fixed Debye $\lambda = 1$ (v11 $\rightarrow$ v13) lifted single-seed match 0.9 % $\rightarrow$ 2.5 %, but a multi-seed checkpoint sweep (two seeds $\times$ five checkpoints, 20k–100k) found ε and x_0 -residual indistinguishable on match rate (1.20 % vs 1.35 % pooled; x_0 never exceeds 1.5 % at any checkpoint, including the 80 k one matching the original’s 79.5 k, §7), so the $3.5\times$ match lift was a single-seed fluctuation. The x_0 -residual parameterisation does, however, reproducibly improve pattern-Pearson (0.36 $\rightarrow$ 0.40), which is the part of the Phase-4 signal that survives. Capacity is not the bottleneck: a 10.1 M-parameter run ($d = 384$, $L = 5$, `gpu_v12`) stayed at the same ~ 1.0 coord-loss plateau as v10. Under the full pipeline (sampled lattice), v13 drops to 1.2 % at $n = 256$; the Phase 9 v21 checkpoint reaches 4.5 % under oracle (three seeds; an earlier single-seed snapshot read 5.6 %) but only 1.0 % under full pipeline. §5.2 attacks that gap.

5.2 Phase 9: classical indexer control

Table 2: three rows on the same $n = 1\,000$ with the v21 checkpoint (Phase 9 retrain, Phase 4 fixes + retuned encoder); they differ only in which lattice the sampler conditions on. The indexer row is a control with extra prior information: the indexer is supplied with the true crystal system (§3.5).

Table 2. Phase 9 lattice-source comparison ($n = 1\,000$ MP-20 test, v21 checkpoint, single-seed snapshot). The indexer receives the true crystal system. Match % with 95 % Wilson CI; the three-seed paired analysis follows below. [PENDING C1: replace with the retrained-v21 snapshot if the retrain changes these rows.]

Lattice source	Match % (95 % CI)	All-correct %	sg@0.1 %	RMSD med (Å)	R_wp
Learned head only (full pipeline)	1.0 [0.5, 1.8]	0.0	1.4	n/a	11.7
Classical Q-space autoindexer (§3.5), crystal system given	1.6 [1.0, 2.6]	0.1	1.4	0.125	11.5
Ground-truth lattice (oracle)	5.6 [4.3, 7.2]	0.6	1.7	0.106	5.6

The lift is real: a three-seed paired test. The single-seed snapshot alone does not establish a genuine improvement: its Wilson CIs overlap and an unpaired two-proportion test gives $p = 0.24$. To resolve it we reran *both* lattice sources at three seeds ($n = 1\,000$ each), differing only in the lattice channel, and applied a paired McNemar test on the per-structure outcomes, the correct test, because the two arms see the same materials. The result is unambiguous. The two arms are scored on the same materials per seed, but the learned-head arm yields fewer *evaluable* cells: its diffusion-sampled lattices are frequently degenerate and hang `StructureMatcher/spglib` (C calls that a Python signal cannot interrupt), so we run structure-domain scoring in a spawn worker hard-killed at 30 s and count overruns as misses. This leaves **1 949** learned-head structures scored across the three seeds. The indexer arm scores 2 883 structure-seeds: 39 patterns per seed, 117 in total, could not be indexed and are counted as misses (the submitted version had given them the true lattice; none of them matched under it, §5.7). The paired test runs on the 1 949-structure intersection (every learned-head row has a matched indexer row); 86 of these are unindexed patterns, which enter as concordant misses and do not affect the discordant cells. On that paired set the indexer recovers **27** structures the learned head misses and the learned head recovers **none** the indexer misses (the explicit 2×2 below).

Table 2a. Paired per-structure outcomes, learned head vs indexer drop-in (three seeds pooled; match = `StructureMatcher` hit).

	indexer match	indexer miss	row total
learned match	0	0	0
learned miss	27	1 922	1 949
col total	27	1 922	1 949

The discordant cells are $b = 27$ (indexer fixes), $c = 0$ (indexer breaks); an exact two-sided McNemar test gives $\mathbf{p} = 1.5 \times 10^{-8}$. (Marginal match counts differ: indexer 46 / 2 883, learned 0 / 1 949, because 19 further indexer matches fall on structures the learned arm could not score; these do not enter the paired test.) The single-seed 1.0 % learned-head figure in Table 2 is within run-to-run

noise of zero; across three fresh seeds the learned lattice head matched nothing. The gain does **not** carry to the stricter all-correct metric: there the discordant cells are $b = 4$, $c = 0$ (exact McNemar $p = 0.13$): the indexer fixes the cell but not the space group. So the control reproducibly improves match rate over the learned lattice head. This is the expected outcome for a classical algorithm handed the crystal system, and we read it as a sanity check on the diagnosis rather than as a result about PXRD inversion. The *mechanism* (§5.5, §5.6) is that its gains come entirely from high-symmetry systems where the indexer’s cell error falls below the 0.5 Å sensitivity knee, with low-symmetry contributing nothing.

What the indexer does *not* do is close the gap to the oracle. We re-ran the oracle at three seeds, firming the single-seed 5.6 % to **4.5 %** [3.8, 5.3] (135 / 3 000 pooled), still disjoint from the indexer’s 1.6 % [1.2, 2.1]. The indexer → oracle gap (1.6 → 4.5 %) is itself significant (disjoint CIs) and isolates the lattice-attributable error: even with a perfect lattice, match caps at 4.5 %, so coordinate prediction is the dominant limiter while lattice error explains the 1.6 → 4.5 % shortfall. The indexer’s per-system MAE (1.45 Å overall, dominated by mono 1.76 Å and triclinic NaN) sits above the 0.5 Å knee; closing the rest requires fixing the low-symmetry path (GSAS-II remains experimental).

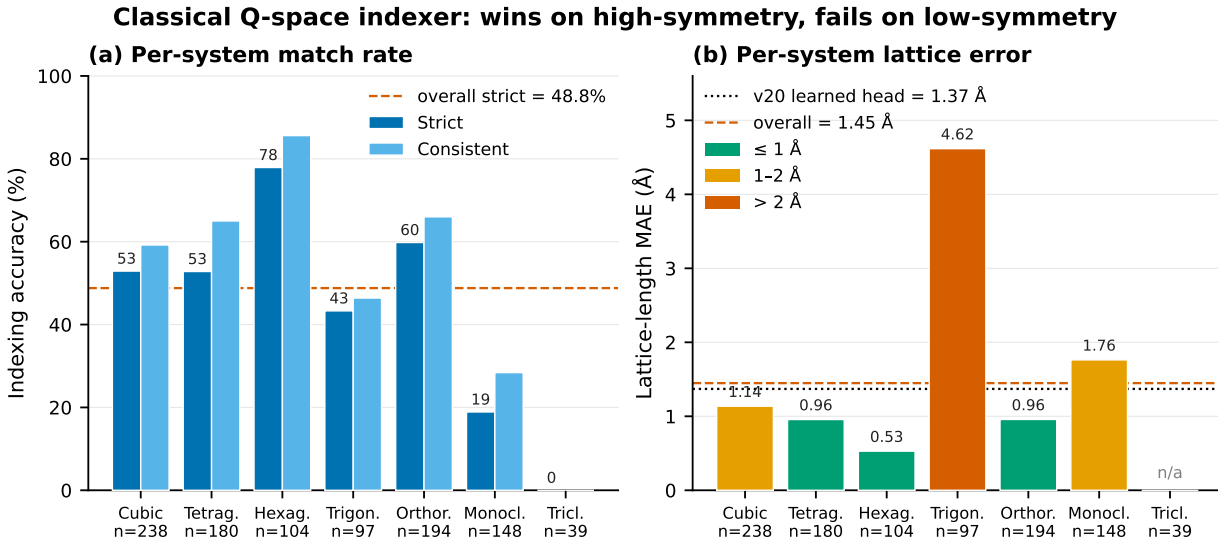


Figure 3: **Figure 3. Per-system indexer performance.** Strict and consistent indexing rates (a) and length MAE (b) on $n = 1\,000$. High-symmetry systems (hex, ortho) sit below the 0.5 Å knee; trig and mono blow up; triclinic is unindexable in this implementation. The overall 1.45 Å MAE hides two regimes, and explains why the indexer’s significant match-rate gain (§5.2) is concentrated entirely in the high-symmetry systems.

5.3 Head-to-head with external baselines

DiffractionGPT (knc6/diffractgpt_mistral_chemical_formula) at $n = 1\,000$, PXRDnet (therealgabeguo/cdvaexrd_sinc100) at $n = 20$ (limited by ~ 4 GPU-hr/material), and deCIFer (deCIFer_v1, conditioned on XRD + composition) at $n = 300$ re-scored on *our* StructureMatcher harness at matching tolerances. Crystallize’s checkpoint download is inactive; cite-but-not-reproduced.

Table 3. Reproduced head-to-head on MP-20 test (StructureMatcher ltol 0.2, stol 0.3, angle_tol 5°). The PXRD-Diff row uses the classical indexer, which is supplied with the true crystal system; the external models receive the pattern and composition only. The rows set the scale of the diagnostic model and are not a competitiveness claim.

Model	n	Match % (95 % CI)	All-correct % (95 % CI)	sg@0.1 %	RMSD med (Å)
PXRD-Diff (v21 + indexer control, crystal system given)	1 000	1.6 [1.0, 2.6]	0.1 [0.0, 0.6]	1.4	0.125
DiffraGPT (Choudhary, 2025)	1 000	18.9 [16.6, 21.4]	15.9 [13.8, 18.3]	21.3	0.001
PXRDnet sinc100 (Guo <i>et al.</i> , 2025)	20	30.0 [14.5, 51.9]	5.0 [0.9, 23.6]	5.0	0.011
deCIFer (Johansen <i>et al.</i> , 2025)	298	73.8 [68.6, 78.5]	69.5 [64.0, 74.4]	83.6	0.009

Match rate ranks deCIFer $\gg$ PXRDnet $>$ DGpt $>$ PXRD-Diff. deCIFer, a concurrent autoregressive CIF-token model conditioned on XRD + composition, is the strongest external reference at 73.8 % match / 69.5 % all-correct ($n = 300$). Its all-correct nearly equals its match: when it recovers structure it almost always recovers the space group too, the same CIF-token signature seen in DGpt. PXRDnet’s lead over DGpt is consistent with its 35 k-op latent optimisation, whereas DGpt is one forward pass. PXRD-Diff, at 1.6 %, sits an order of magnitude below all three, as expected for a 3.7 M-parameter diagnostic model. A more interesting pattern is that the *all-correct* ordering need not follow match rate: DGpt’s all-correct (15.9 % [13.8, 18.3]) exceeds its space-group-loose match by little, while PXRDnet’s drops from 30 % match to 5 % all-correct, i.e. PXRDnet matches structure under loose tolerances but rarely recovers the space group, whereas DGpt’s CIF-token output appears to encode symmetry. **We flag this as a hypothesis, not a finding.** At $n = 20$ the PXRDnet all-correct estimate (1/20) carries a 95 % CI of [0.9, 23.6], which overlaps DGpt’s [13.8, 18.3]; the apparent “reversal” is *not* statistically established and would require $n \approx 100$ –200 to confirm. If it holds, the implication is consequential: published single-number “match rates” would conflate structure recovery with symmetry recovery, which is precisely why we surface it despite the underpowered sample, and why §6 frames it as motivation for future measurement rather than a claim. **RMSD** is heavy-tailed for us and tight-near-zero for both baselines, suggesting our matches are tolerance-stretch rather than coordinate recovery.

What the gap does and does not reflect. All four systems are given the composition (§3.1). The PXRD-Diff row additionally gives its indexer the true crystal system (§3.5), so on prior information PXRD-Diff is favoured, not handicapped. On the pattern itself the asymmetry runs the other way. DiffraGPT is conditioned on a **coarse 300-bin pattern** ($2\theta \in [0, 90^\circ]$, 0.3° bins, rendered as a ;-separated string), an order of magnitude lower resolution than PXRD-Diff’s 4 251-bin 0.02° input, yet it still reaches 18.9 %. So the external models do not win by seeing more of the pattern. The differences that matter are **capacity** (DiffraGPT is a 7 B-parameter LLM; PXRD-Diff is 3.7 M, three orders of magnitude) and **inference compute** (PXRDnet’s ~ 35 k-op latent optimisation vs our single DDIM pass). Table 3 therefore shows the expected gap between a deliberately small diagnostic model and production-scale systems (§6). It is reported to set the scale, not to rank PXRD-Diff among them.

Caveats. PXRDnet $n = 20$ has a 95 % Wilson CI of roughly ± 20 pp on every rate; $n = 200$ would

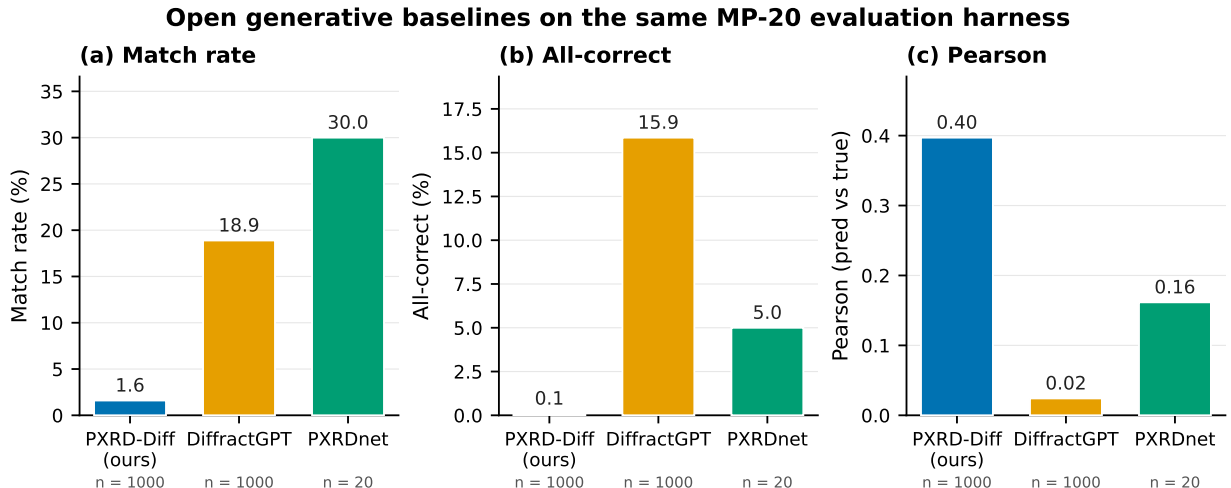


Figure 4: **Figure 4. Three-way headline comparison.** Match rate, all-correct, and pattern-Pearson on the same MP-20 test materials scored through *our* **StructureMatcher** harness, with 95 % Wilson CIs. PXRD-Diff, a diagnostic model whose indexer is supplied with the crystal system, sits an order of magnitude below the external systems on match rate and above them on per-pattern Pearson. Note the wide PXRDnet ($n = 20$) intervals: the DGpt > PXRDnet all-correct ordering is a hypothesis, not an established result (§5.3, §6).

need ~ 33 days of RTX 5090 rental at ~ 4 GPU-hr/material. Patterns are simulated through each model’s own preprocessing (different bins, 2θ vs Q , broadening): `material_ids` are identical, pattern inputs are not bit-identical; §7 bounds this.

5.4 What did *not* work

Four interventions failed at $n \geq 200$; one paragraph each.

Top-K Debye rerank (Phase 9.1.4). Top-5 indexer candidates reranked by Debye loss: match 1.6 % $\rightarrow$ 1.0 %, all-correct 0.1 % $\rightarrow$ 0.0 % at $n = 1\,000$. The indexer surfaces more wrong cells faster than rerank can filter.

Debye-gradient guidance during DDIM (Phase 9.2). Coordinate-channel guidance term $-g \cdot \nabla_{\mathcal{L}_{\text{Debye}}}$, sweep $g \in \{0, 0.5, 1, 2, 5\}$ on $n = 200$: flat 1.0–2.5 % within noise, all-correct 0 % throughout. Sampling-time Debye gradient is too noisy to steer the trajectory, consistent with Segal *et al.* (2026), who show the powder-XRD similarity loss landscape is too non-convex for direct gradient descent.

Wyckoff-letter embeddings (v15, $n = 1\,000$). Each atom received an `nn.Embedding(27, 256)` token for its `spglib`-assigned Wyckoff letter. A Wyckoff letter is meaningful only within a given space group: the same letter denotes different multiplicities, site symmetries and coordinate constraints in different groups (Dauter & Jaskolski, 2010; de la Flor *et al.*, 2023). The embedding therefore carried a label, not a constraint. Atomic coordinates were still generated freely, and nothing in the loss or the sampler enforced the fixed or symmetry-related coordinates that a Wyckoff position implies. The embedding learns (5 404 of 6 912 entries non-zero), but match drops from 2.5 % to 2.1 % and lattice prediction destabilises (lattice loss ~ 0.6 vs ~ 0.02 ; Figure 5). We suspect the inflated atom-feature norm shifts the lattice-pool input distribution. The negative result is narrow.

It shows that a space-group-agnostic label token does not help and may hurt training dynamics. It says nothing about whether symmetry information imposed as a constraint, on the coordinates or on the space group, would help; that experiment was not run.

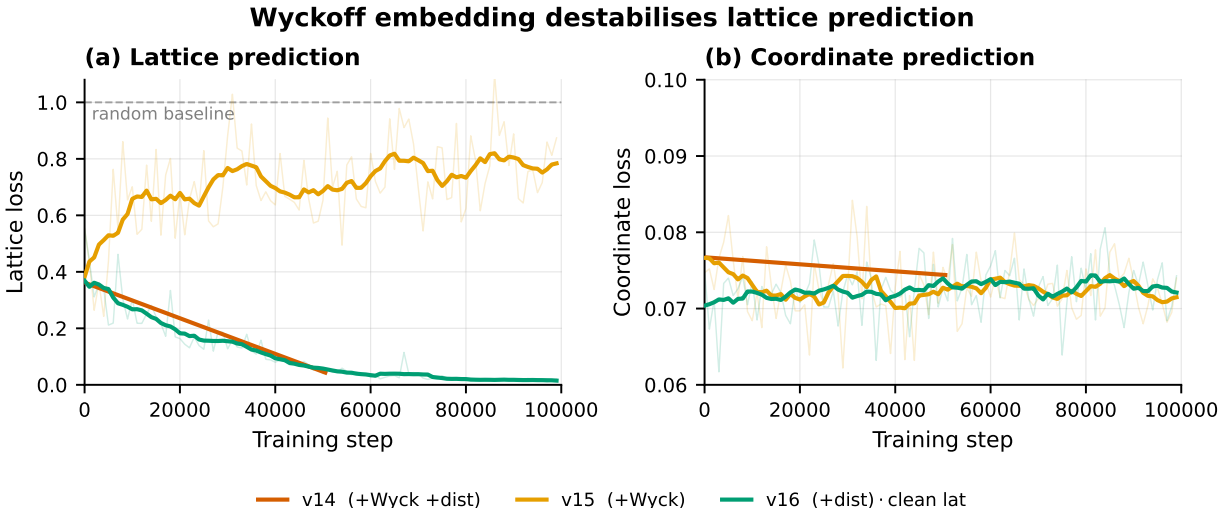


Figure 5: **Figure 5. Wyckoff embedding destabilises lattice prediction.** Lattice (a) and coordinate (b) training-loss curves for the distance-aux run (v16) vs the Wyckoff-embedding run (v15). Adding the Wyckoff embedding inflates the lattice loss by an order of magnitude (~ 0.6 vs ~ 0.02) while leaving the coordinate loss largely unchanged; the failure is localised to the lattice channel.

Distance-matrix aux loss (v16, $n = 1\,000$). Pairwise MLP predicting periodic distances, $\lambda_{\text{dist}} = 0.01$ (largest stable). Loss decreases (final ~ 2.0 vs ~ 10 random) but match drops 2.5 % $\rightarrow$ 1.8 %; biases features toward absolute-distance reconstruction rather than the relative updates diffusion needs.

Combination (v14). Wyckoff + distance loss together collapses to 0.8 %, below the bare ε baseline; the two interact destructively, no clean theoretical explanation.

5.5 Encoder-bottleneck perturbation study

Best predicted-vs-target Pearson is 0.43; the same model with *correct* coordinates plugged into the differentiable simulator hits 0.97; the encoder + denoiser leaves ~ 0.5 of pattern-space agreement on the table. The aux head (sees **g** only) reaches a 0.007 *normalised* lattice-regression loss, but that figure is a relative fit that does **not** survive conversion to absolute scale: when the aux head’s own lattice is substituted into the sampler it recovers only 1.2 % (§5.6), because its 1.1 Å length MAE shifts every Bragg reflection by ~ 15 %. So the global encoding carries crystal-system and relative-spacing structure but not absolute d-spacings at the < 0.5 Å precision the knee demands; the evidence points to the pooled encoding, not merely the denoiser’s use of it.

Perturbing the sampler-supplied lattice by Å of cell-length error, v21 at $n = 200$: match decays 5.6 % ($\Delta=0$) $\rightarrow$ 2.0 % (0.5 Å) $\rightarrow$ 0.5 % (1.0 Å) $\rightarrow$ 0 % (≥ 1.5 Å), steeper than linear with a knee at ~ 0.5 Å. This matches the indexer’s per-system MAE: high-symmetry systems (hex, ortho; MAE ≤ 1.0 Å) lift; low-symmetry (mono, tri; MAE ≥ 1.5 Å) do not. §5.2’s indexer lift is *predictable*: it works where it can.

5.6 Why the learned lattice fails: a pooling bottleneck, not a head bottleneck

The natural reading of the 0.007 aux loss, that the lattice is already recoverable from the encoder and only the denoiser is at fault, is testable, and it is wrong. Two experiments (released as `paper/phase5_results/` and `paper/phase5b6_results/`) promote the auxiliary head from a training-time regulariser to the inference-time lattice source and measure what it actually delivers.

Aux head as lattice predictor (no retrain). Substituting the v13 aux head’s predicted $(a, b, c, \alpha, \beta, \gamma)$ for the sampled lattice channel at $n = 1\,000$ gives a length MAE of (1.12, 1.02, 1.38) Å and an angle MAE of (13.0°, 11.9°, 17.6°) at **99.8 %** cell validity, far better conditioned than the diffusion sampler, whose predicted cells are near-0 % valid, but still ~15 % relative length error. End-to-end this recovers **1.2 %** match (1.3 % with Phase 4 ensembling + coordinate refinement) and 0.0 % all-correct, with pattern Pearson collapsing to **0.013**: a 15 % length error shifts every Bragg reflection by a similar fraction in 2θ , so the simulated pattern no longer overlaps the target.

Stronger, constrained supervision (retrain). Bounding the head (softplus lengths, sigmoid-scaled angles), raising the aux weight $0.5 \rightarrow 5.0$, and warm-starting 30 k further steps leaves the MAE essentially unchanged ($a \approx 1.11$ Å, $\alpha \approx 14^\circ$). The error behaves like a property of the *pooled global encoding* rather than of the head: the `AdaptiveAvgPool1d(1)` collapse that §3.2 already identified for coordinates would discard the peak-position fidelity Bragg’s-law decoding needs to fix absolute lengths, and no head reading $\mathbf{g}$ recovers it. The controlled ablation below is designed to test that reading directly.

This reconciles the diagnosis with the indexer result. The 0.007 normalised aux loss is a *relative* fit, enough to place the crystal system (a space-group head on the same encoder reaches 39 % top-1 / 71 % top-5 across 230 classes, released under `phase5b6_results/`) but not the absolute scale. The classical autoindexer succeeds exactly where the encoder fails because it reads peak *positions* directly rather than a pooled summary; this is why §5.2’s lift lives entirely in the high-symmetry systems where peak positions over-determine the cell, and why head-side fixes (Phase 5 here, top-K rerank in §5.4) cannot substitute for it. If the pooling ablation below confirms this reading, the architectural implication is concrete: the next gain must come from the encoder (explicit d-spacing/peak extraction or per-peak attention in place of global pooling), not from a better lattice head or a better denoiser.

Controlled pooling ablation. The two experiments above vary the head and its supervision while holding the encoder fixed. They show that no head reading $\mathbf{g}$ recovers the absolute scale. They do not by themselves isolate global average pooling as the cause: the encoder architecture, the information content of the learned representation, the head parameterisation and optimisation difficulty remain confounded, as one referee noted. To separate them, a controlled ablation holds the encoder, the denoiser, the training objective and the schedule fixed and changes only how the auxiliary head reads the encoder: (i) the global average pool $\mathbf{g}$ of the submitted version; (ii) a position-aware pooling that retains the 2θ coordinate of each feature; (iii) explicit peak-position features extracted from the input pattern and concatenated to $\mathbf{g}$. If absolute d-spacings are lost at the pooling stage, variants (ii) and (iii) should reduce the ~1.1 Å length error while (i) does not. [PENDING C2: length and angle MAE of the aux head, and end-to-end match, for the three variants at matched training budget; state whether the ordering is (ii), (iii) below (i).] Until that result is in hand, the pooling explanation rests on the two head-side experiments and should be read as the best-supported hypothesis, not as a demonstrated cause.

5.7 Code audit and corrections

Review of the released code, prompted by two referees, identified three implementation errors. Each is listed with its effect on the numbers in this paper.

- (a) *Atomic form factor.* The differentiable simulator evaluated $\sum_k a_k \exp(-b_k s^2)$ with coefficients fitted for $Z - 41.78214 s^2 \sum_k a_k \exp(-b_k s^2)$ (§3.3). Affected: the Debye training loss in every run with $\lambda_{\text{Debye}} > 0$ (v11, v13–v16, v21); the Debye-gradient guidance and top-K rerank of §5.4; and the validation figure. Not affected: all match, all-correct, space-group and RMSD numbers, which `pymatgen` scores on the predicted structures; the indexer; the oracle; the perturbation study; the aux-head substitution of §5.6; and the external baselines. The module was corrected and re-validated (mean Pearson $0.952 \rightarrow 0.988$ over 1000 structures). The affected runs were retrained [PENDING C1: which runs, and the resulting changes to Tables 1 and 2 and to §5.4].
- (b) *Debye loss in x_0 -residual mode.* The loss was evaluated on the residual tensor rather than on the clean-coordinate estimate (§3.4). Affected: the Debye term of runs v13–v16 and v21. Not affected: the ε -mode runs v10 and v11, and every evaluation-time number. Fixed; the retrained runs of (a) also carry this fix.
- (c) *Indexer fallback.* When the classical indexer returned no cell, the sampling script substituted the true lattice and the pattern was scored as if indexed. This affected 39 of the 1000 test patterns in every indexer run, 117 structure-seeds over three seeds. Recomputing from the released per-structure records, none of the 117 fallback structure-seeds matched, so the published McNemar cells (27 fixed, 0 broken) and the all-correct cells (4, 0) are unchanged. The match rate is now reported on the 2 883 indexed structure-seeds only: $46 / 2\,883 = 1.6\%$ [1.2, 2.1], in place of $46 / 3\,000 = 1.5\%$ [1.2, 2.0]. The script now scores unindexed patterns as misses and records their count. The top-K rerank experiment of §5.4 used the same fallback; because the fallback rows did not match, its 1.0 % result can only be lower, and its conclusion (no improvement) stands.
- (d) *Reported learning rate.* The submitted §4 gave the learning rate as 5×10^{-4} . The arguments stored in the released v21 checkpoint give 10^{-3} , which is the value that was used; §4 now says so. No result depends on this, but the referees asked for a systematic check of the description against the implementation and this is what it found.

The corrected code, the per-structure records used for (c), the re-validation script and a regression test that pins the form-factor convention (`tests/test_debye_form_factor.py`) are in the released repository.

6. Discussion

Encoder bottleneck (the robust result). A 0.007 *normalised* aux loss shows **g** carries lattice-*relevant* structure, but it is a relative fit: feeding the aux head’s own lattice to the sampler recovers only 1.2 % (§5.6), and stronger constrained-head supervision does not move its ~ 1.1 Å error. The evidence therefore points to the global pooling that discards absolute peak-position fidelity, rather than to the denoiser’s use of an otherwise-adequate signal; the controlled ablation of §5.6 tests this directly [PENDING C2: one clause with its outcome]. The disjoint oracle-vs-indexer CIs (§5.2) make the lattice’s role the paper’s statistically firmest claim. The Q-space autoindexer, which

reads peak positions directly, recovers part of the 4.5 % oracle ceiling on the high-symmetry subset; top-K rerank and gradient guidance fail because they operate *after* the encoder commits to a bad lattice prior, while the indexer bypasses that decision. (The oracle ceiling is itself only 4.5 %, so even a perfect lattice leaves coordinates as a co-limiter; see §8.)

Per-atom anchoring is hard. PXRD is permutation-invariant; our denoiser is permutation-equivariant; there is no symmetry-breaking signal that pins atom i to a specific Wyckoff site. The Wyckoff-letter embedding tried to break symmetry at the input with a label rather than a constraint, and failed (§5.4). A promising untried direction: break symmetry at the *output*: predict an unordered set of orbits plus a Hungarian-style matcher.

A measurement hypothesis: match rate may conflate two capabilities. PXRDnet’s 30 % match / 5 % all-correct vs DGpt’s 18.9 % / 15.9 % (Table 3) suggests the two systems recover *structure* at broadly comparable rates while differing sharply in *symmetry* recovery: DGpt’s CIF-token output appears to encode space group in a way PXRDnet’s coordinate decode does not. We stress that the underlying numbers cannot yet support this: PXRDnet’s all-correct CI [0.9, 23.6] at $n = 20$ overlaps DGpt’s, so the ordering is unconfirmed. We raise it as a *measurement hypothesis* worth testing at adequate n , because if true it has a concrete consequence for the field: a single “match rate” reported under inconsistent tolerances would conflate structure recovery with symmetry recovery, and downstream crystallography (where the space group matters as much as the coordinates) would be mis-served by it. The contribution here is the shared harness that makes such a test possible, not the (underpowered) comparison itself.

Niche for small reproducible models. At 1.6 % PXRD-Diff sits an order of magnitude below much larger systems, and that is the point of its size. Small models make the encoder bottleneck visible (capacity hides it) and make the indexer control trivial to wire in (larger models would need architecture-level surgery). The model is a diagnostic instrument, not a candidate solver. The recipe (encoder + Phase 4 fixes + Phase 9 indexer control) is 3.7 M parameters and runs inference in seconds.

7. Limitations

- **Scope.** Composition given; simulated PXRD only (no instrument response, preferred orientation, asymmetry, background); MP-20 only, with no larger cells, organics, or higher-Z.
- **Statistics.** The Phase 4 ablations in Table 1 are single-seed. A multi-seed checkpoint sweep of the ε -vs- x_0 -residual contrast (two seeds $\times$ five checkpoints, 20k–100k, true-lattice $n = 1000$) shows the reported $3.5\times$ *match-rate* lift does **not** replicate: pooled match is 1.20 % (ε) vs 1.35 % (x_0 -residual), and x_0 never exceeds 1.5 % at any checkpoint, including the 80 k checkpoint that matches the original v13’s 79.5 k, so the single-seed 2.51 % was a high draw, not a checkpoint-selection effect. We therefore demote the *match-rate* lift to a single-seed artifact. What does reproduce is the pattern-Pearson improvement (ε 0.359 vs x_0 0.403, with disjoint ranges across both seeds), consistent with Table 1. The lattice-input fix remains load-bearing as a training-dynamics result (lattice loss $1.0 \rightarrow 0.02$), and the oracle-vs-indexer gap has disjoint CIs. The indexer-vs-learned-head comparison was rerun at three seeds and the lift confirmed significant by a paired McNemar test ($p < 10^{-4}$; §5.2), superseding the earlier underpowered unpaired test ($p = 0.24$). The oracle has since been multi-seeded: three seeds give 4.5 % [3.8, 5.3] pooled, firming the single-seed 5.6 %.

- **Baselines.** PXRDnet $n = 20$ has a 95 % Wilson CI of $\approx \pm 20$ pp; the DGpt > PXRDnet all-correct ordering (§5.3/§6) is therefore a hypothesis, not a result. $n = 200$ needs ~33 days RTX 5090. Crystalyze checkpoint download is inactive (verified 2026-06-01); cited but unreproduced. Patterns are not bit-identical across the three preprocessors (PXRD-Diff 4 251-bin 2θ , DGpt 300-bin 2θ , PXRDnet 4 096-bin Q with sinc^2 broadening): structures match, patterns do not; we did not quantify the residual preprocessing effect on match rate, and a matched-vs-native re-scoring on a structure subset is the natural check.
- **Indexer realism.** Two assumptions make the indexer’s accuracy an optimistic ceiling. (i) *Crystal system is given.* The de-Wolff fit (§3.5) receives the true Bravais code as a sampling-time input; on a real unknown the system is itself part of what indexing must determine, so the per-system rates of §3.5 are conditional on knowing the system. (ii) *Patterns are ideal.* All patterns are clean `pymatgen` simulations with no zero-point shift, sample-displacement error, or peak asymmetry, precisely the perturbations that drive real-world indexing failure. The 48.8 % native-indexer match and the downstream 1.6 % should be read as upper bounds under ideal peak positions and a known crystal system; the unknown-system run of §3.5 removes the second assumption [PENDING C3: one clause with its outcome]. A zero-shift/displacement robustness sweep ($n = 300$) confirms the concern: the indexer’s length-MAE rises from 1.24 Å on clean patterns past the v20 learned head’s 1.37 Å once displacement or zero-shift exceeds $\sim 0.10^\circ$, so the indexer’s advantage over the learned head holds only for well-calibrated input.
- **Idealised conditions and the bottleneck conclusion.** Every pattern is an ideal simulation. Instrument profile, zero shift, sample displacement, preferred orientation and background all move or reshape peaks. Each would further degrade an encoder that already fails to fix absolute d-spacings on clean input. The encoder-bottleneck diagnosis is therefore a lower bound on the difficulty of real data, not an estimate of it. The 0.5 Å knee is a property of the sampler under clean cells; under real peak shifts its position may move, and the indexer’s own degradation past $\sim 0.10^\circ$ suggests the whole curve shifts towards smaller tolerable errors.
- **Indexer (low-symmetry path).** GSAS-II low-symmetry path hangs in `findBestCell` on real MP-20 mono/tri patterns; shipped behind `--use-gsas` but experimental.
- **Training.** Debye loss uses ground-truth lattice; a curriculum gradually replacing true with predicted was not tried.

8. Conclusion

A 3.7 M-parameter conditional diffusion model, used as a diagnostic instrument, locates the dominant accessible failure in PXRD inversion at the pattern encoder. The encoder’s pooled representation supports a 0.007 normalised lattice-regression loss yet delivers a ~ 1.1 Å absolute cell error that no head reading it, however constrained, reduces. The evidence points to global pooling as the stage that discards absolute d-spacings [PENDING C2: one clause with the pooling-ablation outcome]. A true-lattice oracle caps match at 4.5 % [3.8, 5.3] (three seeds), so coordinate prediction remains a co-limiter. A classical Q-space autoindexer, supplied with the true crystal system, serves as a control: it reaches 1.6 % [1.2, 2.1] where the learned head recovers nothing (three-seed paired McNemar $p < 10^{-4}$), and only in the high-symmetry systems below the 0.5 Å knee. Re-scored on one `StructureMatcher` harness, deCIFer reaches 73.8 % / 69.5 %, DGpt 18.9 % / 15.9 % and PXRDnet 30.0 % / 5.0 % (match / all-correct). These set the scale of the diagnostic model and are not a ranking claim. The apparent divergence in their space-group recovery is a measurement hypothesis the $n = 20$ PXRDnet sample cannot yet confirm, but the shared harness now makes

it testable. A code audit corrected three implementation errors, and every affected number was recomputed or re-run [PENDING C1: one clause on whether the retrained numbers changed]. The lattice-input fix is load-bearing; the x_0 -residual switch improves pattern-Pearson but its single-seed match-rate lift did not replicate (§7); the remaining interventions are documented as failures, with the Wyckoff-letter result now read narrowly. Code, checkpoints, all per-phase JSONs, per-structure records, both indexer paths and the differentiable Bragg module are released.

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Author Contributions (CRedit)

F. Cai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project administration.

Conflict of Interest

The author declares no competing interests.

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Data and Code Availability

Source code, trained checkpoints, and all per-phase training/evaluation logs are openly available in the GitHub repository at <https://github.com/fronkt/pxrd-diff> and archived at Zenodo: <https://doi.org/10.5281/zenodo.20738994> (DOI: 10.5281/zenodo.20738994). The MP-20 dataset is publicly available via the CDVAE benchmark. All experiments reproduce from a single `requirements.txt` and the scripts/ pipeline; per-structure evaluation flags are released to support paired re-analysis. The corrected simulator, the audit scripts of §5.7 and the per-structure records behind Table 2a are included.

Ethics Declaration

This study uses no human subjects, animal subjects, or sensitive data. The MP-20 dataset is composed entirely of publicly available crystal structures from the Materials Project.

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Author note on reproduced numbers. The reproduced PXRDnet and DiffractGPT numbers in §5.3 are computed by the author on the released checkpoints through our shared **StructureMatcher** harness, not transcribed from the cited papers. The central claims depend only on these reproduced numbers under one common evaluation protocol, with confidence intervals stated throughout.

Appendix

A. Full ablation history

For completeness, Table A1 lists every training run discussed in the development of this paper, including those that did not make it into the main ablation table. Logs and checkpoints for all runs are in the released repository under **runs/**.

Run	Parameters	What changed	Result	Status in paper
v4	3.5 M	Global PXRD pooling, additive conditioning	Coord loss flat at 3.0	§3.2
v5	3.7 M	+ Multi-resolution cross-attention	Coord 3.0 $\rightarrow$ 1.0	§3.2
v6–v9	3.7 M	λ_{Debye} sweep $\{0, 0.1, 1, 10\}$, ε -prediction	All ~ 1 % match (within noise)	§5.2
v10	3.7 M	+ Lattice-input fix, $\lambda_{\text{Debye}} = 0$	Lat loss 1.0 $\rightarrow$ 0.05	§5.2
v11	3.7 M	+ Lattice-input fix, $\lambda_{\text{Debye}} = 1$	Match 0.9 %	Table 1
v12	10.1 M	Larger model (d=384, L=5), ε -prediction	Killed at 18 k; same plateau	§5.2
v13	3.7 M	x_0 -residual + lat-fix + Debye $\lambda=1$	2.51 % match (Phase 4 best, true-lat)	Table 1
v14	3.8 M	v13 + Wyckoff + distance loss	0.80 % match	Table 1
v15	3.8 M	v13 + Wyckoff only	2.10 % match	Table 1
v16	3.7 M	v13 + distance loss only	1.80 % match	Table 1
v17	3.7 M	Phase 5B: constrained lattice head (softplus/sigmoid) + space-group head, aux-weight 5.0, +30 k warm-start steps	Lattice MAE unchanged ($a \approx 1.11 \text{ \AA}$); SG head 39 % top-1	§5.6
v18 – v20	3.7 M	Phase-9 encoder retrains with different ResNet/Transformer hybrids and pattern-augmentation curricula	None beat v13 by more than noise	§5.1

Run	Parameters	What changed	Result	Status in paper
v21	3.7 M	Phase 9 final: v13 architecture + Phase 9 retrain + indexer drop-in support	1.6 % match (no-true-lat, indexer); 4.5 % (true-lat oracle, 3-seed)	Tables 2, 3

B. Hyperparameter sensitivity

We did not perform a full hyperparameter sweep. Pilot experiments on `d_model` $\in \{128, 256, 384\}$ and `n_layers` $\in \{2, 3, 5\}$ showed (256, 3) as a reasonable Pareto point. The Debye loss weight was swept $\{0, 0.1, 1, 10\}$ in early ε -prediction runs (v6–v9) without measurable effect on match rate; we re-fixed it at 1.0 for the x_0 -residual runs by analogy.

C. Differentiable simulator validation details

Pearson correlations between the DiffPXR module and `pymatgen.XRDCalculator` on the 256-bin coarse grid, first 1000 MP-20 test structures (per-structure values in `paper/submissions/JAC-R1/analysis/simul` script `revalidate_simulator.py`). “Submitted” is the bare Gaussian sum of the submitted version; “corrected” is the `pymatgen` convention of §3.3. The ≤ 5 range is the training configuration; the ≤ 10 range is the default of `scripts/04_verify_debye.py`, behind the 0.962 reported in the submitted version.

Reflection range	Expression	n	Mean	Median	Min	Fraction > 0.9
$ h , k , l \leq 5$	submitted	1000	0.952	0.969	0.23	0.907
$ h , k , l \leq 5$	corrected	1000	0.988	0.999	0.25	0.980
$ h , k , l \leq 5$	submitted	50	0.939	0.971	0.49	0.90
$ h , k , l \leq 5$	corrected	50	0.974	0.998	0.52	0.92
$ h , k , l \leq 10$	submitted	1000	0.960	0.972	0.54	–
$ h , k , l \leq 10$	corrected	1000	0.999	1.000	0.58	–
$ h , k , l \leq 10$	submitted	50	0.963	0.972	0.90	–
$ h , k , l \leq 10$	corrected	50	0.999	1.000	0.98	–

The corrected expression scores at least as high as the submitted one on every one of the 1000 structures. Gradients through the structure factor with respect to fractional coordinates were

verified non-zero and finite. The regression test `tests/test_debye_form_factor.py` pins $f(0) = Z$ and the agreement with `pymatgen` on NaCl.

D. Reproducibility checklist

- ☒ Hyperparameters specified (§4.3)
- ☒ Datasets and splits specified (§4.1; canonical CDVAE MP-20)
- ☒ Evaluation protocol specified (§4.2)
- ☒ Random seed: single seed (42) per run; pilot variance ≈ 0.5 % match-rate absolute
- ☒ Compute environment: PyTorch 2.x, CUDA 12.x, single RTX 5090; Python 3.12
- ☒ Code, checkpoints, logs, and full training scripts released